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RESEARCH

Advancing forest fire prediction: A multi-layer stacking ensemble model approach

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Abstract
A reliable forest fire probability map is vital for disaster management and an essential resource in land use planning. This study evaluates the efficacy of the multi-layer stacking ensemble Machine Learning (ML) method for forest fire susceptibility mapping, presenting a comparative case study within the Malakand division of Pakistan. Our extensive literature review shows that the present ML model has never been used in Pakistan's forest fire scenarios. We employed several benchmark models for comparative evaluation, including Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), Extreme Gradient Boosting (XGBoost), and K-Nearest Neighbor (KNN). A comprehensive fire inventory database was constructed, including satellite and ground hotspot data and relevant influencing factors. The fire probability indices from the six models were analyzed and validated using accuracy, area under the curve (AUC), precision, recall, and F1 score evaluation metrics. According to the Performance Evaluation Outcomes, the multi-layer stacking ensemble model provides the best outcomes in terms of accuracy 96.24%, AUC 99.43%, precision 97.81%, recall 94.59%, and F1 96.17% metrics. These results underscore the model's potential as an effective new forest fire Probability mapping tool. Given its evidenced effectiveness, local forestry authorities in the Malakand division should consider its application for enhanced forestry conservation management and fire prevention strategies.

Keywords Forest fire · Multi-layer ensemble model · Machine learning · Probability mapping · Malakand division

Introduction

Forest fires are a global environmental concern and are particularly destructive in Pakistan's Malakand Division. Forests, rich in biodiversity (Köhl et al. 2015; Morales-Hidalgo et al. 2015), play a vital role in conserving water and soil and regulating carbon cycles (Piao ShiLong et al. 2019; Sun et al. 2023; Pan et al. 2023; Anees et al. 2024d; Khan et al.

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2020; Muhammad et al. 2023; Haider et al. 2017; Jallat et al. 2021). Fires disturb forest ecosystems, harming biodiversity, altering species composition, and disrupting structures. They also impact natural resources, economies, and communities (Meddour-Sahar 2015; Rowell and Moore 2000; Shahzad et al. 2024; Mehmood et al. 2024a). The increase in frequency and intensity of forest fires globally over the past decade has raised concerns about their environmental and socio-economic effects, including global warming and threats to the survival of various species (Singh et al. 2021; Tariq et al. 2021; Gao et al. 2024). Because of these risks, early fire prediction and detection measures are essential in decreasing the destruction and resources expended on fighting the fire (Abid 2021). Among the critical steps in forest fire management, mapping fire susceptibility, detecting hotspots, monitoring active fires, and assessing post-fire degradation (Adab et al. 2013), mapping susceptibility is crucial for identifying fire-prone areas and aiding decision-making in fire management and prevention (Eugenio et al. 2016).

Technological advances in remote sensing have revolutionized fire monitoring and evaluation over the years (Gitas et al. 2012), driving the discovery of ecosystem changes (Miller and Thode 2007; Anees et al. 2024b; Mehmood et al. 2024c) and more practical land use and cover mapping (Anees et al. 2025; Topaloğlu et al. 2016; Hussain et al. 2024). Satellite images are essential elements used in wildfire management (Vega-García and Chuvieco 2006; Zhou et al. 2024a, b), mapping burn severity and thus enhancing the need for coordinated fire protection (Syifa et al. 2020; Wu et al. 2015). Forest fires are driven by climatology, vegetative, topographical characteristics, demographics, and economic variations that vary in temporal and spatial platforms (Guo et al. 2016; Ma et al. 2020; Morgan et al. 2001). In Pakistan, the dry sub-tropical climate exacerbates the vulnerability of sub-tropical and temperate forests to fires, with altitudinal variations influencing fire spread at different elevations (Saeed et al. 2016; Mehmood et al. 2024b). Every year, a discernible portion of the country's forests are burned in the sub-tropical and moist temperate regions due to fires that may be natural, self-ignition due to the accumulation of dry leaves, deforestation, and accident-collision (Mehmood et al. 2024d). According to (Hansen et al. 2013), over 80% of these fires are attributed to anthropogenic activities. Although biotic and abiotic factors causing the heterogeneous density and frequency distribution of fires are well understood, Pakistan does not have operational plans for active fire management at regional and national levels This underscores the urgent need to propose and implement effective fire management strategies (Mannan et al. 2017).

Scholars have developed various ML algorithms to predict wildfire occurrences and estimate burned areas. Applications include spread control (Botequim et al. 2017; Zhai et al. 2020), damage assessment (Botequim et al. 2017), event prediction, resource deployment, and technological support (Duff and Tolhurst 2015; Marques et al. 2012). A 2019 survey reviewed 300 significant publications, identifying the most frequently used ML techniques across various domains. It discussed prominent algorithms in supervised, unsupervised, and agent-based frameworks and included essential ML references (Jain et al. 2020a). Indeed, from LR, RF, AdaBoost, XGBoost, Artificial Neural Networks (ANNs), SVMs, KNN, and many others, the methods used are diverse (Jain et al. 2020; Anees et al. 2024a, c; Zhou et al. 2024c). Nevertheless, a standard limitation is the regional focus of most studies, with analyses typically confined to specific areas rather than global datasets, including locales like Margalla Hills in Pakistan (Tariq et al. 2022), Alberta in Canada (Vega-Garcia et al. 1996), Galicia in Spain (Alonso-Betanzos et al. 2003), Lesvos in Greece (Vasilakos et al. 2007), Lebanon (Sakr et al. 2011), Turkey (Özbayoğlu and Bozer 2012), Australia (Dutta et al. 2016), Daxinganling Mountains in China (Chen et al. 2015), Kroumirie Mountains in Tunisia (Toujani et al. 2018), Chile (Bjånes et al. 2021), and the South Central US (Wang and Wang 2019). Typically, single ML models come with a range of shortcomings. Ensemble learning integrates multiple individual models into a unified framework to enhance predictive accuracy and generalization, effectively addressing the limitations of single ML models (Li et al. 2023; Luo et al. 2024). While ensemble learning and deep learning methods have achieved considerable success across domains (Y. Xie and Peng 2019), their application remains underexplored in specific decision-making contexts, such as forest fire management and natural resource optimization, especially in developing regions like Pakistan (Johnson et al. 2017). This study introduces a novel multi-layer stacking ensemble approach, combining RF, SVM, and XGBoost models, to predict forest fire susceptibility. Additionally, the integration of Standard Deviation Ellipse analysis provides a unique spatial evaluation of fire dynamics, offering insights into high-risk zones. Such a methodology has not been previously applied to forest fire prediction in Pakistan, highlighting its potential to fill critical gaps in the field.

This study addresses these gaps by applying single and ensemble ML models to predict forest fire occurrences in Pakistan's Malakand division. Specifically, the study aims to evaluate the performance of single ML models against multi-layer stacking ensemble models in predicting forest fire incidents. Additionally, spatial analysis techniques, such as standard deviation ellipses and centroid displacement, were used to identify high-risk fire areas and assess the historical evolution of fires in the region. The primary objectives of this study are: (1) to compare the efficiency of single ML models and ensemble models in predicting forest fire occurrences; (2) to identify the optimal combination

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of ML models for fire risk assessment; (3) to apply spatial analysis techniques to detect high-risk areas in the Malakand division; and (4) to develop generalizable predictive models that can be applied to similar regions worldwide. By filling the research gap in forest fire prediction in Pakistan, this study aims to contribute to developing more effective fire management strategies in vulnerable regions.

Materials and methods

Study area

The research focused on the Malakand division, covering the period from 2001 to 2023. The study area, Malakand division, is geographically located in the western zone of south Asia, and its longitude and latitude range from 71.430 South to 73.850 North and from 36.070 West to 36.400 East. Malakand division is Positioned in the northwest of Khyber Pakhtunkhwa province and shares boundaries with Gilgit Baltistan, district Battagram and Kohistan of Hazara division to the east, Afghanistan lies to the west, Nooristan district of Afghanistan to the north, and the districts of Charsadda and Mardan to the south. Malakand division has nine districts: Malakand, lower Dir, upper Dir, lower Chitral, upper Chitral, Swat Shangla, Buner, and Bajaur. The division covers an area of 32,000 km^2, and the forest covers 8 km^2. Malakand division recorded 612 fire events from 2001 to 2023, as shown in Fig. 1.

Malakand division has a variety of landscapes ranging from the mountains in the north to the sparsely populated areas in the southwest. It has four seasons, including a cold and dry winter in (Dec to Feb), a dry and hot spring from March to May, a rainy from (Jun to Aug) and a cycle of monsoons between (Sep and Nov) (Begum et al. 2011; Shaw 2015). Pakistan's forest cover stands at merely 4.5%, which is a significant concern given the nation's agriculture-based economy and its location within the South Asian Ecological Zone (Oliveira et al. 2011). During the latter half of the 20th century, there has been increasing evidence of a rise in wildfire occurrences across the country, leading to an expansion of burned areas (Rafaqat et al. 2022). Comprehensive

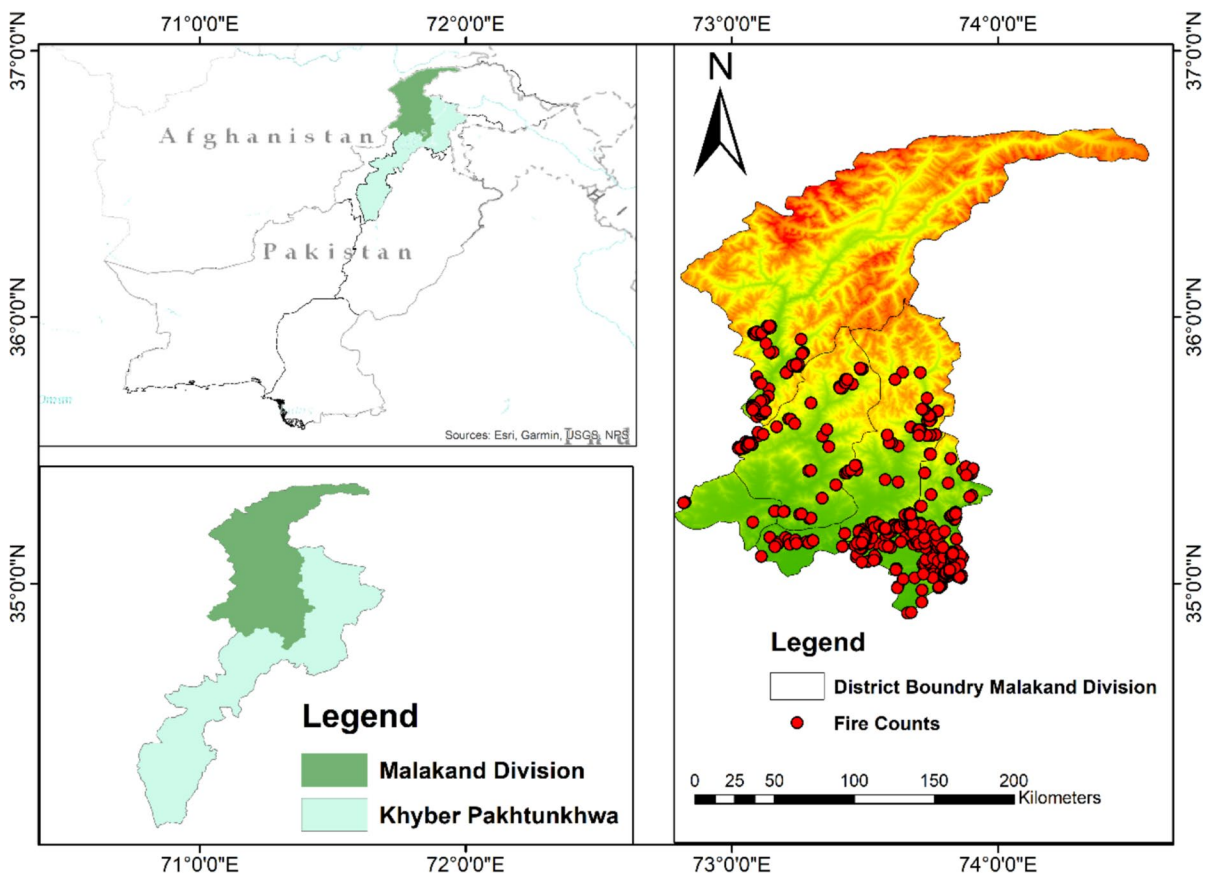

Fig. 1 Map of the study area showing the Malakand Division within Khyber Pakhtunkhwa, Pakistan. The inset maps illustrate the location of Khyber Pakhtunkhwa province in Pakistan (bottom-left) and Malakand Division within the province (top-left). The main map highlights the elevation profile of Malakand Division (green to red gradient) and fire incident locations (red points) recorded during the study period. The division boundary is marked for spatial reference

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national wildfire research is vital, considering this environmental concern, particularly the availability of remote sensing technologies and worldwide databases that provide more detailed comprehension of factors causing fires and enhanced prediction models (Rafaqat, Iqbal, Kanwal, & Son-Rafaqat et al. 2022a, b; Cai et al. 2024).

Handling of response variable

The method encompasses two primary data sources for the study of the forest dynamics in the Malakand Division, Khyber Pakhtunkhwa, Pakistan. The primary dataset spans from 2014 to 2023, sourced from local forestry records. To fill the gap from 2001 to 2013, the spatial distribution of fire pixel data from the MODIS Collection 6 sensor was employed, and it was recognized for its reliability in monitoring vegetation fires. This MODIS data, obtained from the FIRMS fire archive, includes information on fire timing, location, duration, perimeter, scale, direction, and speed of spread, along with confidence values ranging from 1–100%(Chuvieco et al. 2008; Mupfiga et al. 2022; Vadrevu et al. 2012). Only fire pixel data with confidence levels of 80% or higher were included in the analysis to ensure reliability. The datasets were aligned spatially and temporally, and GIS analysis, using shapefiles in the Geographic WGS84 projection, was employed to extract fire characteristics. Limitations, such as potential biases and ethical considerations, were addressed, acknowledging the challenges and implications of integrating datasets from different periods and sources. The land use cover data was taken from the European Space Agency (ESA) World Cover 2020 at 10 m resolution-based Sentinel-2 data. The World Cover product comes with eleven land cover classes. It has been generated in the ESA World Cover project framework, part of the ESA's 5th Earth Observation Envelope Program (EOEP-5) (Zanaga et al. 2021).

Selection and handling of predictor variables

A balanced dataset was generated with random points representing fire and non-fire incidents, assigning binary thresholds of 1 for fire and 0 for non-fire incidents. The random point selection process was guided by prior experience to minimize data overdispersion. Following established practices, a 1:1 ratio of fire to non-fire points was maintained throughout (Denham et al. 2008). This selection adhered to the principle of ensuring randomness in both spatial and temporal dimensions, as emphasized by (Ma et al. 2020; Mehmood et al. 2024e). In generating these random points, we involved R Studio and ArcGIS 10.8 software and placed these points within the forested areas. To accurately define the range of forest land, we relied on ESA land use data (https://developers.google.com/), accessed on 1st January 2024, from the year 2020. This method ensured that our non-fire points were only created within these regions, excluding areas such as rivers or urban lands (Pang et al. 2022). We achieved a total of 1212 fires and no fire points.

In the present study, we categorized the independent variables into four key groups based on their relevance to fire ignition: vegetation, climate, anthropogenic, and topography. All the specific variables considered for each primary category were selected with consideration of current established knowledge of what triggers fire ignition, thus assuring that the analysis only captured preceding elements with an identified influence on fire occurrence (Milanović et al. 2022; Nhongo et al. 2019; Ye et al. 2017). Remote sensing vegetation indices are used for the vegetation category, as shown in Supplementary Material Figure S1. For example, EVI and NDVI were used to measure the quantity and quality of vegetation cover through spectral observations expressed as the vegetation state or moisture content (Barreto and Armenteras 2020). These indices were sourced from the MOD13Q1.061 Terra Vegetation Indices dataset and given at a 250 m pixel level (Didan 2021).

Additionally, the Leaf Area Index (LAI) and Fraction of Photosynthetically Active Radiation (FPAR) indices were obtained from the MOD15A2H V6.1 MODIS dataset, are given at a 500 m pixel level (Myneni et al. 2021). The tree cover was accessed from the MOD44B Version 6.1 Vegetation Continuous Fields yearly product, and they also have global coverage with a 250 m resolution (DiMiceli et al. 2022). As shown in Table 1, the calculation methods of these indices can be used to build a foundation for analyzing the role of vegetation in forest fires.

Monthly historical climatic data in our research were obtained from two different data repositories: the ERA5 climate reanalysis data downloaded from the Copernicus Climate Change Service site (Zhang et al. 2021) and TerraClimate data as shown in S2, accessed via the database *("IDAHO_EPSCOR/TERRACLIMATE")* on January 1, 2024. The list of variables abstracted from these repositories is presented in Table 2.

However, human activities play a vital role in fires. One reliable indicator for human activities used in this work is the population density, as shown in Supplementary Material

Table 1 Calculation methods of indices

Vegetation Index	Universal Formula
NDVI	NIR – RED / NIR + RED
EVI	2.5 × (NIR – RED / NIR + 6 × RED – 7.5 × BLUE + 1)
LAI	Total leaf area (m^2) / Ground area (m^2)
FPAR	a × NDVI + b

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Table 2 Descriptions of independent variables

Category	Independent variable	abbreviation	source	unit	resolution	time
vegetation	Normalized Difference Vegetation Index	NDVI	MODIS/061/MOD13Q1	nm	250 m	2001–2023
	Enhanced Vegetation Index	EVI				
	Leaf area index	LAI	MODIS/061/MOD15A2H	M	500 m	2001–2023
	Fraction of Photosynthetically Active Radiation	FPAR				
	Tree Cover	TC	MOD44B Version 6.1	%	250 m	2001–2023
Climatic	minimum temperature	Tmin	*IDAHO_EPSCOR/TERRACLIMATE*	°C	4638.3 m	2001–2022
	maximum temperature	Tmix		°C		
	wind speed	WS		m/s		
	vapor pressure	VP		kPa		
	actual evapotranspiration	AE		mm		
	Total precipitation	Ppt	ERA 5 climate reanalysis data	mm	9 km	2001–2023
	soil temperature	Tsoil		°C		
	Forecast albedo	AB		%		
	surface net solar radiation	NSR		J m^{-2}		
	surface net thermal radiation	NTR		J m^{-2}		
	total evaporation	E		M		
	skin temperature	ST		°C		
	surface pressure	SP		Pa		
anthropogenic	Population density	Pop	https://hub.worldpop.org/	count	1 km	2001–2020
	distance from the road	disRO	https://www.openstreetmap.org	M	*1:2,372,791*	2015
	distance from the river	disRI		M		
	distance from the buildup	disBU		M		
topographic	Slope	S	https://earthexplorer.usgs.gov/	M	30 m	2020
	Aspect	A				
	Elevation	E				
	hill shade	HS				
	topographic witness index	TWI				

figure S3, obtained from the WorldPop population dataset (Song et al. 2005). Other measures obtained from OpenStreetMap are distances to roads, rivers, and built-up areas (Hu et al. 2016). Topographical information extracted from the DEM of the study area includes the Topographic Wetness Index, slope, elevation, aspect, and hillside, as shown in Supplementary Material Figure S4. These measures are crucial in understanding the physical components that can influence the likelihood and spread of fires. Average TWI, elevation, slope, aspect, and hill shade of all defined areas were obtained in a 1 × 1 km grid using the GIS software ArcGIS 10.8. The method of generating TWI used in this work is detailed in (Xie et al. 2022) (Eq. 1), further showing the detailed analytical methodological considerations considered in this study.

$$TWI = \ln\left(\frac{SCA}{tan\ slope}\right) \tag{1}$$

Methodological overview of statistical and ML models

Standard deviation ellipse (SDE)

The SDE method is utilized to characterize the spatial distribution attributes of a study object. This technique allows for a visual interpretation of various properties of the object under study, including centrality, distribution range, orientation, and geometric shape (Lefever 1926). This study's method enables precise measurements of the central gravity, distribution scope, direction, and alterations in forest fires across Malakand division. The mathematical formula for the SDE is the following (Jing et al. 2023):

Center of gravity is (Eq. 2):

$$\bar{X} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}, \bar{Y} = \frac{\sum_{i=1}^{n} w_i y_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

Directional angle is (Eq. 3)

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$$\tan\theta = \frac{A+B}{C} \tag{3}$$

where A, B and C are (Eqs. 4, 5, 6):

$$A = \sum_{i=1}^{n} w_i^2 \tilde{x_i}^2 - \sum_{i=1}^{n} w_i^2 \tilde{y_i}^2 \tag{4}$$

$$B = \sqrt{\left(\sum_{i=1}^{n} w_i^2 \tilde{x_i}^2 - \sum_{i=1}^{n} w_i^2 \tilde{y_i}^2\right)^2 + 4\sum_{i=1}^{n} w_i^2 \tilde{x_i}\tilde{y_i}} \tag{5}$$

$$C = \sum_{i=1}^{n} 2 w_i^2 \tilde{x_i}\tilde{y_i} \tag{6}$$

Standard deviation of X and Y axis are (Eq. 7 and 8):

$$\delta_x = \sqrt{\frac{2\sum_{i=1}^{n}\left(w_i \tilde{x_i}\cos\theta - w_i \tilde{y_i}\sin\theta\right)^2}{\sum_{i=1}^{n} w_i^2}} \tag{7}$$

$$\delta_y = \sqrt{\frac{2\sum_{i=1}^{n}\left(w_i \tilde{x_i}\sin\theta - w_i \tilde{y_i}\cos\theta\right)^2}{\sum_{i=1}^{n} w_i^2}} \tag{8}$$

where n is the number of variables. $x_i y_i$ are the Geographical coordinates (longitude and latitude) of each variable. w_i is a Weight representing the forest fire metric for each variable. $\bar{X}, \bar{Y}$ is a Weighted average coordinate of the centroid. θ is the Orientation angle of the ellipse. δ_x, δ_y is a Standard deviation of the ellipse along the X and Y axes, respectively. $\tilde{x_i}\tilde{y_i}$ is a Deviations of the coordinates of each variable from the mean centroid.

Random forest (RF)

The RF model is a comprehensive ensemble method building on decision trees (Chuvieco et al. 2008; Anees et al. 2024a). Ensemble learning can combine many decision trees, improving not only the accuracy but also the robustness of prediction (Guan 2023). The RF model involves creating many classification trees determined by bootstrap samples. Most of the data, about two-thirds, is used to learn the model, and one-third, the out-of-bag (OOB) samples, is used for validation (Duro et al. 2012; Anees et al. 2024c). The output is determined by averaging across all votes for the multiple decision tree predictions (Chen et al. 2021). The general expression of the RF model is described mathematically as follows in (Eq. 9) (Shao et al. 2023):

$$h(x) = \frac{1}{T}\sum_{t=1}^{T} h\left(x, \theta_t\right) \tag{9}$$

The total number of decision trees is represented by T, where each θ_t is a random vector distributed independently and identically, and x is the input vector, the final result is calculated as the average outcome of the regressions from all sub-trees $h\left(x, \theta_t\right)$. The model's prediction performance is affected by the deterministic selection of random features and the number of trees. The most helpful method for statistically evaluating variable importance within an RF model is the OOB method. This method is simply the increase in OOB error if we randomize a particular variable X^j and keep all the other variables fixed (Abdel-Rahman et al. 2013; Marston et al. 2014). According to the equation given by (Liang et al. 2016), we can calculate the importance of a variable (Eq. 10):

$$VI\left(X^j\right) = \frac{1}{ntree}\sum_{t}\left(err' OOB_t^j \, errOOB_t\right) \tag{10}$$

where $ntree$ is the number of trees, $errOOB_t^j$ is the OOB error for t-th tree without alteration in any variables, and $err' OOB_t^j$ is the OOB error for the tree t when we alter X^j. We can use this comparison to determine how our model's prediction performance would depend on the variable; the misclassification probability in classification tasks is merely proxied by OOB error.

Support vector machine (SVM)

SVM is a popular ML algorithm with excellent supervised learning, particularly in a binary classifier (Ireland and Petropoulos 2015; Tien Bui et al. 2016; Mehmood et al. 2024a). Introduced by Cortes and Vapnik, SVM has quickly become an algorithm of interest across many fields, primarily due to persistent sensor technology improvements and data's fast-growing ubiquity. As a result, SVM became a valuable algorithm for solving real-world problems, such as fire prediction, making its way into many studies (Brown et al. 2018; Gigović et al. 2019). The main advantage of SVM is efficient work with nonlinear issues due to kernel functions: SVM can work with nonlinear data differently from most other models. A kernel function enables the model to distribute the data to high-dimensional spaces; this is where separation and classification become relatively easy due to the linear nature of relations. Taking a binary classification SVM as an example, a training set $T = \{(x_1, y_1), \cdots (x_1, y_1)\} \in (X \times Y)^l$, where $x_i \in X = R^n$ and $y_i \in \{1, -1\}$ for $(i = 1, 2, \ldots, l)$ Defines the feature vector. The consequence parameter C and the kernel function K (x, x') are specified. The trouble of optimization is then formulated and resolved in the following manner(Boubeta et al. 2015) (Eqs. 11, 12):

$$\min_{\alpha} \frac{1}{2}\sum_{i=1}^{j}\sum_{j=1}^{1} y_i y_j a_i a_j k\left(\varkappa, x'\right) - \sum_{j=1}^{1}\alpha_j \tag{11}$$

$$s.t. \sum_{i=1}^{j} y_i \alpha_i = 0, 0 \le \alpha_i \le C, i = 1, \ldots, l \tag{12}$$

The optimal solution $\alpha^* = (\alpha^*, \cdots, \alpha^*)^T$ is obtained. A positive component $\alpha^* : 0 \le \alpha_j^* \le C$ is then selected, and the threshold is computed as (Pang et al. 2022) (Eq. 13)

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$$b^* = y_j - \sum_{i=1}^{1} y_i \alpha_i K(x_i - x_j) \tag{13}$$

Finally, the decision function is constructed (Eq. 14):

$$f(x) = sgn(\sum_{i=1}^{1} \alpha_i * y_i K(x, x_i) + b^* \tag{14}$$

Logistic regression (LR)

LR is a valuable statistical method for predicting fire occurrences and assessing the factors contributing to this type of event in different regions of the world (Garcia et al. 1995; Martínez et al. 2009). LR models the connection between a binary outcome variable, where '1' signifies a "Yes" response, and one or more predictor variables (X) using a link function. It includes diverse geographical locations in which it has been used to "identify and quantify the factors that affect fire occurrence (Martell et al. 1987). LR formally presented as an (Eq. 15):

$$P(Y = 1) = \frac{1}{1 + e^{-X\beta}} \tag{15}$$

where P (Y = 1) is the probability of fire. Even though this model integrates multiple predictor variables, as a multiple linear regression model, considering it is used for predicting a binary outcome, it includes a logistic function presented with the following formula (Guo et al. 2016) (Eq. 16)

$$\eta = X\beta = \beta_0 + \sum_{j=1}^{P} \beta_j X_{ji} = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_n X_{in} \tag{16}$$

Here, η is the link function used in linear models, and $\beta_0 - \beta_n$are coefficients of the model that are determined using a LR approach. This function is defined between 0 and 1, which results in an S-shaped curve that shows the probability of fire as a function of multiple predictors.

Extreme gradient boost (XGBoost)

XGBoost, introduced by Chen and Guestrin in 2016, stands out as a groundbreaking gradient-boosting decision tree (GBDT) framework (T. Chen and Guestrin 2016). It advances the field by employing Taylor's second-order expansion for optimizing the loss function, showcasing superior computational efficiency and broader generalization capabilities relative to other ML techniques (Xie et al. 2022). The XGBoost model represents (Eq. 17):

$$\hat{y}_i = \sum_{k=1}^{k} f_k(x_i), f_k \in F \tag{17}$$

Here, $\hat{y}_i$ denotes the predicted value for the ith sample, k represents the total number of decision trees, x_i corresponds to the input data for the ith sample, $f_k(x_i)$ is the kth decision tree produced in the kth repetition, and f_k is an element of the tree collection space F(Shahzad et al., 2024).

The objective function for XGBoost is (Eq. 18):

$$Obj = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \sum_{k=1}^{k} \Omega(f_k) = \sum_{i=1}^{N} l[y_i, \hat{y}_i^{t-1} + f_t(x_i)] + \sum_{k=1}^{k} \Omega(f_k) \tag{18}$$

In Eq. (18), it consists of two primary components: the first term represents the loss function, quantifying the error between the predicted and observed values. The second term serves as a regularization factor, which controls model complexity, optimizes the tree structure, and mitigates the risk of overfitting. (Linfei et al. 2019).

K-nearest neighbors (KNN)

KNN is suitable for regression and classification. In regression mode, it calculates the average value of the attributes of the neighbors and assigns the value to the object. The use of this model in combination with GIS or remote sensing data for the estimation of forest fire or its detection is widely described (Bui et al. 2018; Mittal et al. 2019; Thach et al. 2018). The most crucial parameters in the KNN model are the number of nearest neighbors, the distance metric, and the attributes of those neighbors. In the case of continuous outcomes, the model calculates the target value as in (Eq. 19):

$$y_i = \frac{1}{w_i} \sum_{j=1}^{k} w_{ij} y_{ij} \tag{19}$$

$i$is the target element of interest and j are its neighbors, k is the counter for the neighbors; $w_{ij}$are the weight assigned to the contribution of $h$jth neighbor to the prediction of the hi th element. The weight factor, w_i, is required to be included in a model for factoring the importance of different neighbors based on the distance or other parameters after all of this is used for granulating the estimates of the target element (Bhadoria et al. 2021).

Ensemble model

Stacking is a heterogeneous ensemble learning model (Wolpert 1992), where multiple classifiers with differing methodologies are combined to create a strong classifier that significantly improves the generalized performance (Healey et al. 2018). This study employed a multi-layer stacking model, enhancing traditional approaches by structuring layers like a neural network. Initially, diverse ML models such as LR, RF, SVM, KNN, and XGBoost are trained, and their prediction values are accumulated. These values are then used to create new datasets, which serve as input for training several meta-models at each layer. Typically, we utilized XGBoost for its robustness across diverse data structures. This method repeats through

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multiple layers, where each layer's output is transformed and fed as input to the next, creating a neural network-like cascade of learning. The final predictive output is synthesized after n layers, with the number of layers determined by performance metrics such as accuracy and loss minimization on a validation set. This structured repetition aims to refine accuracy and performance by leveraging variations in training datasets derived from selective prediction values, as illustrated in Fig. 2.

Hyperparameter tuning

Hyperparameter tuning was performed for all models, including the ensemble stacked model, using grid search cross-validation to enhance model performance and reproducibility (Contreras et al. 2021; Kaur et al. 2022). The number of trees (n_estimators) for the RF model varied between 100 and 500, and the maximum depth (*max_depth*) was adjusted from 10 to 50. SVM tuning focused on kernel type (linear, polynomial, or radial basis function), regularization parameter (C) between 0.1 and 10, and kernel coefficient (γ) for RBF kernels within the range of 0.01 to 1. The XGBoost model involved optimizing learning rate (η) from 0.01 to 0.3, *max_depth* from 3 to 10, the *n_estimators* from 100 to 500, subsample ratios and the γ parameter for tree pruning. For the KNN model, the number of neighbors (k) was adjusted between 3 and 20, while the LR model focused on fine-tuning the regularization

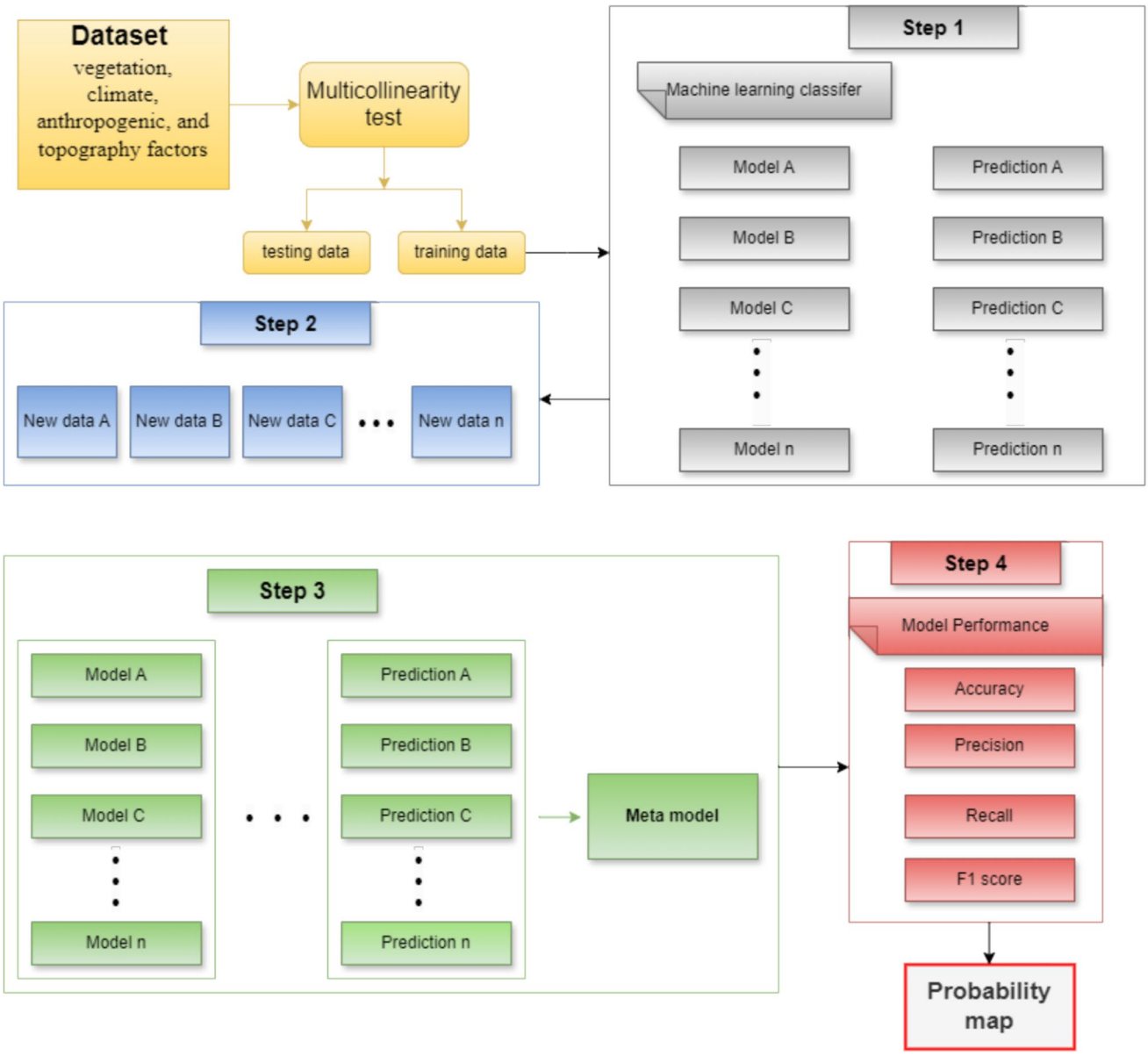

Fig. 2 Flowchart depicting the sequential stages of data processing and corresponding outputs in the multi-layer stacking ensemble framework

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parameter (penalty) with L1 and L2 norms. The ensemble stacked model included RF, SVM, LR, and KNN in the base layer (Sharma et al. 2022). Hyperparameters for each base model were optimized individually using the process described above. Once the base models were tuned, the meta-model (XGBoost) was fine-tuned separately for η, *max_depth*, and the number of boosting rounds. This hierarchical tuning approach ensured that each base model contributed effectively to the ensemble and that the ensemble itself operated at peak performance without overfitting (Dong et al. 2023; Jia et al. 2023). The final parameter configurations for all models were selected based on validation accuracy, using k-fold cross-validation to validate model generalizability (Ramezan et al. 2019). A 10-fold cross-validation approach was applied, dividing the dataset into ten equal parts (Awad and Fraihat 2023). Nine subsets were used for training the model, and the remaining subset was used for testing, iterating this process across all subsets to ensure each data point was used for validation exactly once (Jabbar and Khan 2015). This technique minimizes overfitting and provides a robust estimate of model performance by evaluating it on unseen data. Stratified sampling maintained the class distribution of fire and non-fire points across all folds, ensuring that both classes were well-represented during training and validation (Zaidi 2023).

Quality measures and interaction analysis

For assessing prediction models regarding the likelihood of forest fires, it will also be appropriate to use a confusion matrix, also known as an error matrix, designed as an n×n matrix for comparing forecasting accuracy (Li et al. 2020). To compare models in terms of this likelihood, it is necessary to consider the following criteria: accuracy, precision, AUC, recall, and F1 score (Shao et al. 2022; Badshah et al. 2024). Accuracy is defined as the proportion of true results (both true positives and true negatives) among the total number of cases examined (Eq. 20). Precision is the ability of the classifier to identify only the positive cases; it is the proportion of correct positive predictions to all positive predictions made by the classifier (Eq. 21). Recall is the proportion of cases correctly predicted as positive to the actual number of positive cases (Eq. 22): F1 score is the harmonic mean of precision and recall, which finds a balance between the two metrics (Eq. 23).

$$Accuracy = \frac{TP + TN}{TP+TN+FP+FN} \tag{20}$$

$$Precision = \frac{TP}{TP+FP} \tag{21}$$

$$Recall = \frac{TP}{TP+FN} \tag{22}$$

$$F1 = \frac{(1+a^2)\,PR}{a^2\,(P+R)} \tag{23}$$

The Receiver Operating Characteristic curve ROC proves how well a model predicts an outcome. The AUC can also be used to quantify how successful the model was. The value of AUC spans from 0.5 to 1.0, and higher values show that the model suits well and effectively distinguish classes (Shahzad et al. 2024). To analyze the interaction effects between features, SHAP (SHapley Additive exPlanations) was employed (Iban and Aksu 2024). A SHAP interaction heatmap was generated by computing the pairwise correlations of SHAP values for all features (Nohara et al. 2022; Sun et al. 2023). This visualization technique was used to identify and highlight the relationships and interdependencies between predictor variables, offering enhanced interpretability of the model's predictions.

SHAP-based feature analysis and interaction assessment

SHAP (SHapley Additive exPlanations) was employed to systematically analyze feature importance and evaluate interaction effects among predictors (Kim et al. 2024). This method was selected for its ability to provide both global and local interpretability of the model's predictions, making it particularly effective for complex prediction tasks such as forest fire risk assessment (Cakiroglu et al. 2024; Ponce-Bobadilla et al. 2024; Meng et al. 2023) (Eq. 24).

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left[f(S \cup \{J\}) - F(S) \right] \tag{24}$$

Here, ϕ_j denotes the SHAP value for feature j (predictors), where N represents the set of all features. S refers to a subset of features that excludes feature j. The term $f(S \cup \{J\})$ is the model's prediction when feature j is included, while f(S) is the prediction when feature j is excluded. The factorial components |S|! and (|N|−|S|−1)! serve to weight each subset of features, ensuring that all possible feature interaction combinations are considered (Wang et al. 2022; Zaker Esteghamati et al. 2023).

SHAP values were utilized to rank the importance of predictors based on their mean absolute contributions to the model's predictions (Baptista et al. 2022). This approach enabled the identification of key features that significantly contributed to the overall performance of the predictive model (Nahiduzzaman et al. 2024). The ranking process systematically reduced the initial set of predictors while retaining the most influential features for further analysis. To assess the interactions among predictors, SHAP interaction values were computed, capturing the pairwise relationships between features (Moncada-Torres et al. 2021). A SHAP interaction heatmap was generated to visualize

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these interdependencies and highlight how predictors interact to influence model outcomes jointly (Yu et al. 2024). This visualization facilitated a better understanding of non-linear relationships and the combined effects of features within the predictive framework. The choice of SHAP for this study was driven by its ability to provide consistent, model-agnostic explanations. By attributing contributions to individual features and their interactions, SHAP enhances the interpretability of complex ML models, enabling more informed decision-making in forest fire management (Wikle et al. 2023).

Results

Trends of forest fire

The statistical analysis of monthly and yearly fire count data over 23 years was performed using the Mann-Kendall trend and Sen's slope estimator. The annual results from the Mann-Kendall test show a positive Kendall's tau value of approximately 0.323, with a p-value of about 0.032. Additionally, Sen's slope estimation indicates a median slope of 1.0. This means that, on average, the number of fires has increased by 1 each year over the study period. These results indicate a statistically significant upward trend in the fire counts from 2001 to 2023. The Mann-Kendall test showed no significant trend across the months, as evidenced by a p-value of 0.381. However, Sen's slope analysis suggested a potential average increase of about 2.29 monthly fires, as shown in Fig. 3.

Figure 3(A) shows the temporal patterns in fire occurrences, marked by initial fluctuations, notable increases, and specific periods in 2006, 2009, 2012, 2016, and 2022. While Fig. 3(B) shows the monthly temporal trend of forest fires, there is a specific increase in June, July, and November. These findings stress the importance of targeted interventions and heightened vigilance during peak fire months to mitigate risks and protect communities and ecosystems, underscoring the dynamic and nuanced nature of forest fire trends in the Malakand division.

Multicollinearity and spearman rank correlation analysis

This study initially considered 34 independent variables encompassing vegetation, climate, topography, and anthropogenic factors. A systematic selection process was employed to narrow these down to 16 key features for the final analysis. The first step involved assessing multicollinearity using Variance Inflation Factors (VIF) and Tolerance (TOL) values. Variables with VIF values exceeding 10 or TOL values below 0.1 were excluded, as they indicated high multicollinearity, potentially compromising the statistical reliability of the model. This filtering ensured the remaining variables were statistically independent and suitable for inclusion in predictive modeling, as detailed in S1. Following the multicollinearity assessment, Spearman rank correlation analysis was conducted to evaluate the relationship between each variable and forest fire occurrences while avoiding high inter-correlation among predictors. Variables that exhibited weak or non-significant correlation with fire ignition were eliminated, prioritizing features that demonstrated strong and direct relevance to the dependent variable S5. This dual-filtering approach ensured that the selected variables were independent and highly relevant to forest fire

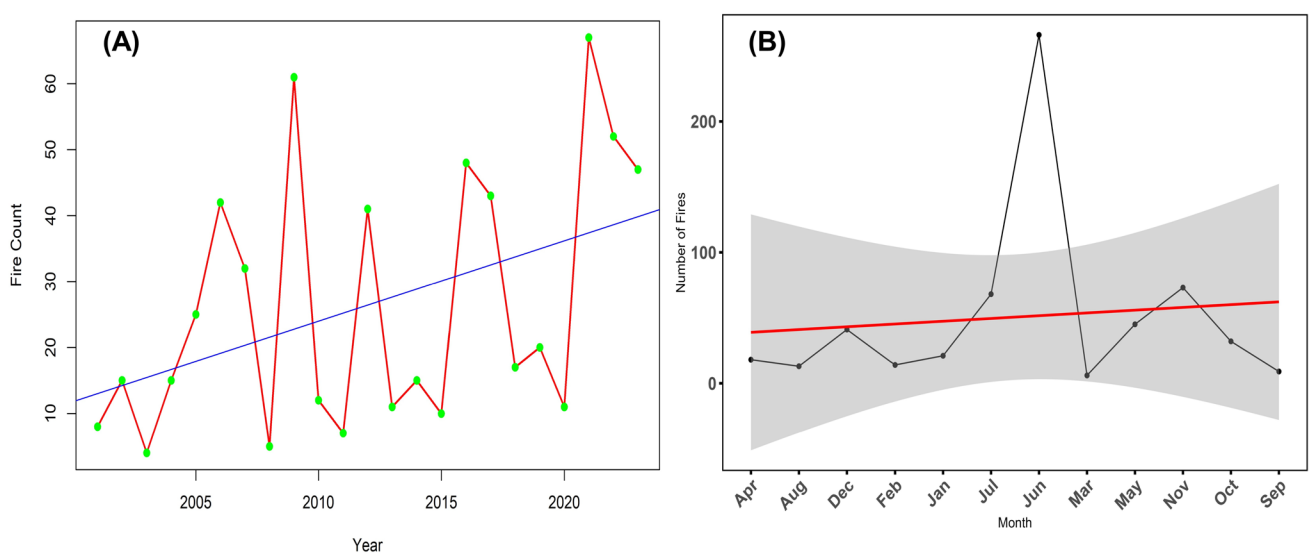

Fig. 3 **A** Temporal trend of annual forest fire occurrences in the Malakand Division from 2001 to 2023. **B** Monthly distribution of forest fires

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prediction. The final set of 16 features, which include critical variables, forms a strong foundation for enhancing the model's predictive accuracy. By eliminating redundancy and ensuring statistical independence, this process strengthens the reliability of the analysis and improves the interpretability of the results.

Standard deviation ellipse

The analysis of forest fire events in Malakand Division, Pakistan, over four distinct periods spanning from 2001 to 2023, offers a detailed view of the evolving spatial and temporal dynamics of these events, as shown in Fig. 4. From 2001 to 2006, the SDE results indicate the central coordinates of fire incidents at approximately 72.37° longitude and 34.74° latitude. The dispersion of these occurrences is about 230.55 m along the X-axis and 687.67 m along the Y-axis, with a Rotation angle suggesting a directional trend in fire spread at 137.06°. This period is characterized by a significant degree of elongation in the fire spread pattern, as denoted by an oblateness value of 0.739, pointing to a specific directional preference or pattern in the spread of fires. As we transition into the subsequent period from 2007 to 2012, there is a noticeable shift in the central coordinates to 72.30° longitude and 34.79° latitude, alongside changes in dispersion and a decrease in the Rotation angle to 129.69°, indicating a shift in the major fire activity's orientation and a less pronounced directional preference, as reflected by the oblateness value decreasing to 0.559.

Progressing to the period between 2013 and 2018, further shifts in central coordinates are observed, now centering around 72.20° longitude and 35.01° latitude, with alterations in dispersion along both axes and a slight increase in the Rotation angle to 135.27°. These changes suggest a nuanced alteration in the directional trends of fire spread, with the oblateness value increasing to 0.696, indicating a return to a more elongated or flattened shape. The final phase from 2019 to 2023 sees another shift in central coordinates to 72.43° longitude and 34.70° latitude, with the dispersion along the X and Y axis adjusting significantly and the Rotation angle slightly decreasing to 134.38°. This last period showcases a decrease in oblateness to 0.525, signifying a relative reduction in the elongation or flattening of the fire spread pattern (Table S2). These comprehensive insights into the spatial and temporal patterns of forest fire occurrences in the Malakand Division over time are crucial for formulating effective fire management strategies, allowing for adaptations to the dynamic nature of fire spread observed throughout these periods.

Performance evaluation of predictive models

The predictive performance of six forest fire incident classification models was evaluated, with results summarized in (Table 3). This analysis highlights the accuracy hierarchy among models and emphasizes the enhanced predictive capacity when combining individual model strengths through a multi-layer stacking ensemble approach. The ensemble model significantly outperformed all individual models, achieving the highest accuracy (96.24%) and F1 score (96.17%). These results underscore their ability to integrate multiple methodologies, leading to more reliable

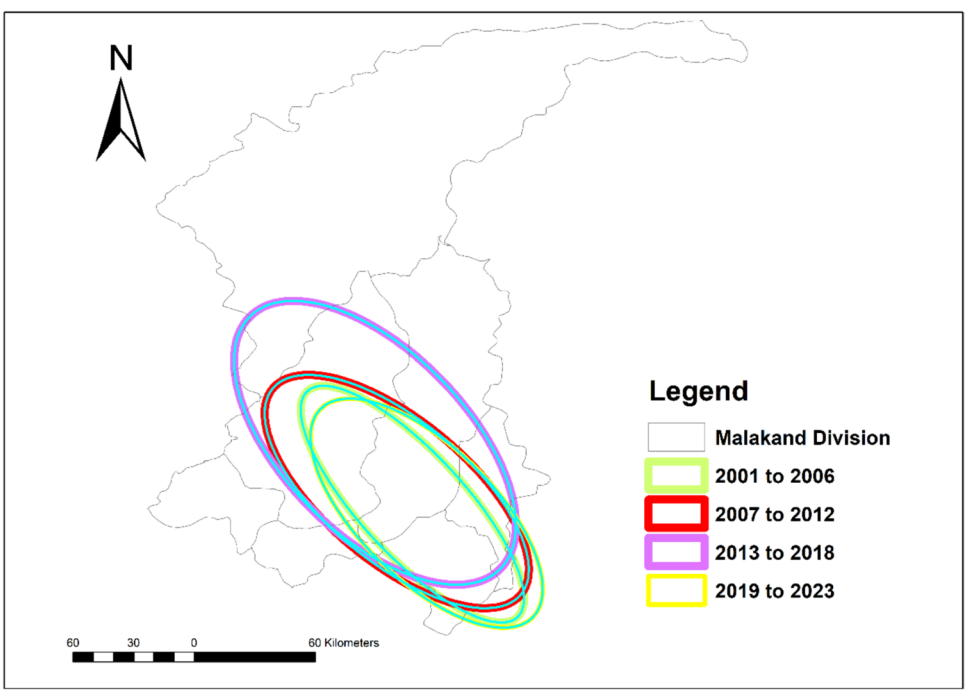

Fig. 4 ESD analysis showing the spatial dispersion and directional trends of forest fires in Malakand Division, Pakistan, across four time periods: 2001–2006 (green), 2007–2012 (red), 2013–2018 (purple), and 2019–2023 (yellow). The ellipses illustrate the mean center, spatial spread, and directional bias of fire occurrences over each period, highlighting shifts in fire hotspots and their spatial dynamics over time

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Table 3 Comparative performance metrics of individual ML models and the multi-layer stacking ensemble model for forest fire prediction

Model	Type	Accuracy	F1 Score	ROC AUC	Recall	Precision
Random Forest	Individual Model	91.16%	91.11%	97.16%	90.61%	91.62%
Extreme gradient boost	Individual Model	88.95%	88.89%	97.07%	88.40%	89.39%
Support Vector Machine	Individual Model	86.46%	85.88%	93.00%	82.32%	89.76%
Logistic Regression	Individual Model	85.36%	84.37%	91.32%	79.01%	90.51%
K-Nearest Neighbors	Individual Model	79.83%	79.08%	88.54%	76.24%	82.14%
Multi-Layer Stacking Ensemble	Ensemble Model	96.24%	96.17%	99.43%	94.59%	97.81%

predictions. Additionally, the ensemble model's ROC AUC of 99.43%i shown in (Fig. 5) exceeded that of any individual model, indicating exceptional capability in distinguishing between fire and non-fire samples. Its recall (94.59%) and precision (97.81%) further acknowledge its efficacy in correctly identifying fire incidents while minimizing false positives.

The individual models also demonstrated varied levels of performance. RF and XGBoost exhibited relatively high accuracy (91.16% and 88.95%, respectively) and robust precision-recall balance. In contrast, SVM and LR showed moderate performance, while KNN had the lowest predictive metrics, highlighting its limitations in handling the complexities of forest fire data. The findings highlight the value of ensemble approaches for complex prediction tasks such as forest fire classification. The integration of diverse model strengths within a structured framework not only enhances accuracy but also provides resilience to data variability. These results support ensemble-based modeling techniques in operational fire management systems, where accurate and timely predictions are critical. RF and XGBoost were the most influential base models, contributing 21.11% and 20.60%, respectively, to the ensemble's overall predictive power (S6). This aligns with their high individual performance metrics, such as Accuracy and F1 Score. In contrast, K-Nearest Neighbors contributed the least, likely due to their relatively lower individual performance. These contributions were calculated using a contribution analysis method based on SHAP values, which quantify the relative importance of each base model in the ensemble. The superior performance of the multi-layer stacking ensemble (Accuracy: 96.24%) indicates that the combination of these diverse base models effectively leverages their strengths, such as the robustness of RF and the gradient-based optimization of Extreme Gradient Boosting, while mitigating individual weaknesses, such as the sensitivity of K-Nearest Neighbors to noise in the data.

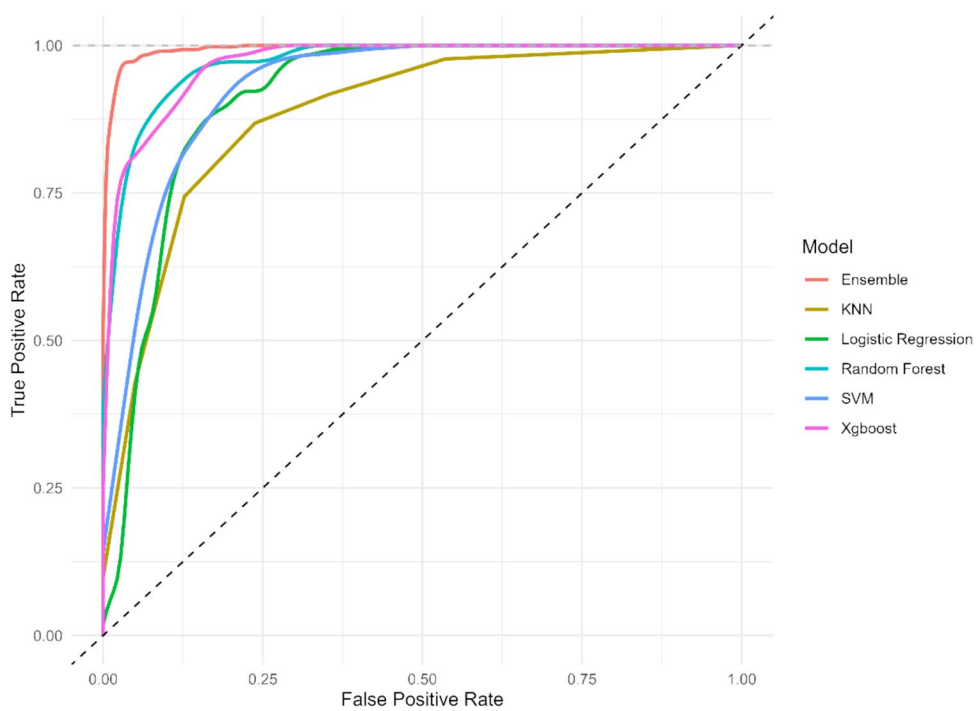

Fig. 5 ROC curves illustrate the AUC values for six models: Ensemble, KNN, LR, RF, SVM, and XGBoost. The ensemble model demonstrates the highest AUC, indicating superior prediction performance compared to the individual models

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Feature interactions and contributions

A SHAP interaction heatmap (Figure S7) was generated to further explore feature interactions and nonlinear impacts. The heatmap highlights the interplay between key variables, such as Precipitation and Tree Cover, which exhibited strong correlations, indicating their combined influence on fire prediction. Similarly, Vapor Pressure and Wind Speed demonstrated significant nonlinear interactions, reinforcing the model's ability to capture critical environmental dynamics relevant to fire risk assessment.

The SHAP analysis evaluated the importance of individual features and their contributions to the predictive model. The variable importance rankings, derived from the mean absolute SHAP values, highlight the relative influence of each feature on the model's predictions. Figure S6 presents the variable importance plot, where Tree Cover, LAI, and Vapor Pressure emerge as the most influential predictors, contributing significantly to the model's performance. These findings align with the ecological understanding that vegetation cover and climatic conditions are critical drivers of fire risk.

The SHAP summary plot (Fig. 6A) provides further insights into the distribution of SHAP values across all predictors. This visualization reveals the variability in feature contributions at the instance level, showcasing how different feature values (e.g., high or low) impact the model's predictions. For example, Tree Cover demonstrates consistently high SHAP values, underscoring its critical role in predicting fire occurrence, whereas features like Hill shade **and** Distance from the Urban exhibit comparatively lower SHAP values, indicating less influence on the model's output. These analyses confirm the selection of the most important predictors for the final model and demonstrate the model's ability to capture the nonlinear relationships between variables and fire risk.

Spatial distribution of fire risk

The multi-layer stacking model, which achieved the highest precision among the six predictive models, was used to map forest fire probabilities across the Malakand Division. The resulting probability map categorizes fire risk into three classes: low probability (0–0.4), medium probability (0.4–0.8), and high probability (0.8–1). (Fig. 7) highlights areas with the highest fire probabilities, primarily located in the forests of Shangla District, including Alpuri, Puran, Chakesar, Martung, and Besham. Significant high-risk zones are also observed in Buner District, particularly in Karakar, Daggar, Ambela, and Malndrai. Additional hotspots are identified in Upper and Lower Dir's mountainous regions and Swat District, particularly in Kalam, Malam Jabba, and Mayadam. Other notable risk zones are found in the forested regions of Malakand, Bajaur, and Lower Chitral. This probability map offers valuable insights into the spatial distribution of fire risk, helping to prioritize fire prevention and

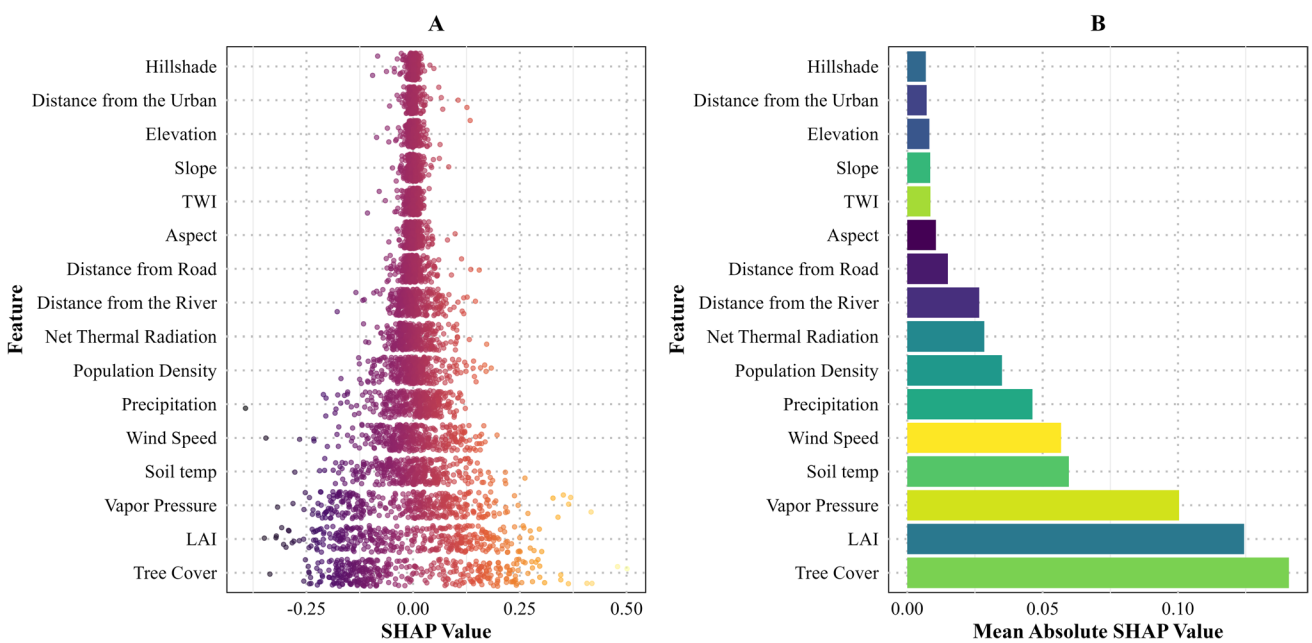

Fig. 6 **A** SHAP summary plot: This plot visualizes the distribution of SHAP values for each feature, highlighting their contributions to model predictions. Each point represents a single prediction, and its position indicates the feature value and SHAP impact. **B** Feature Importance plot: The bar chart ranks features based on their mean absolute SHAP values, illustrating their overall contributions to the predictive model. Tree cover, LAI, and vapor pressure are the most influential features

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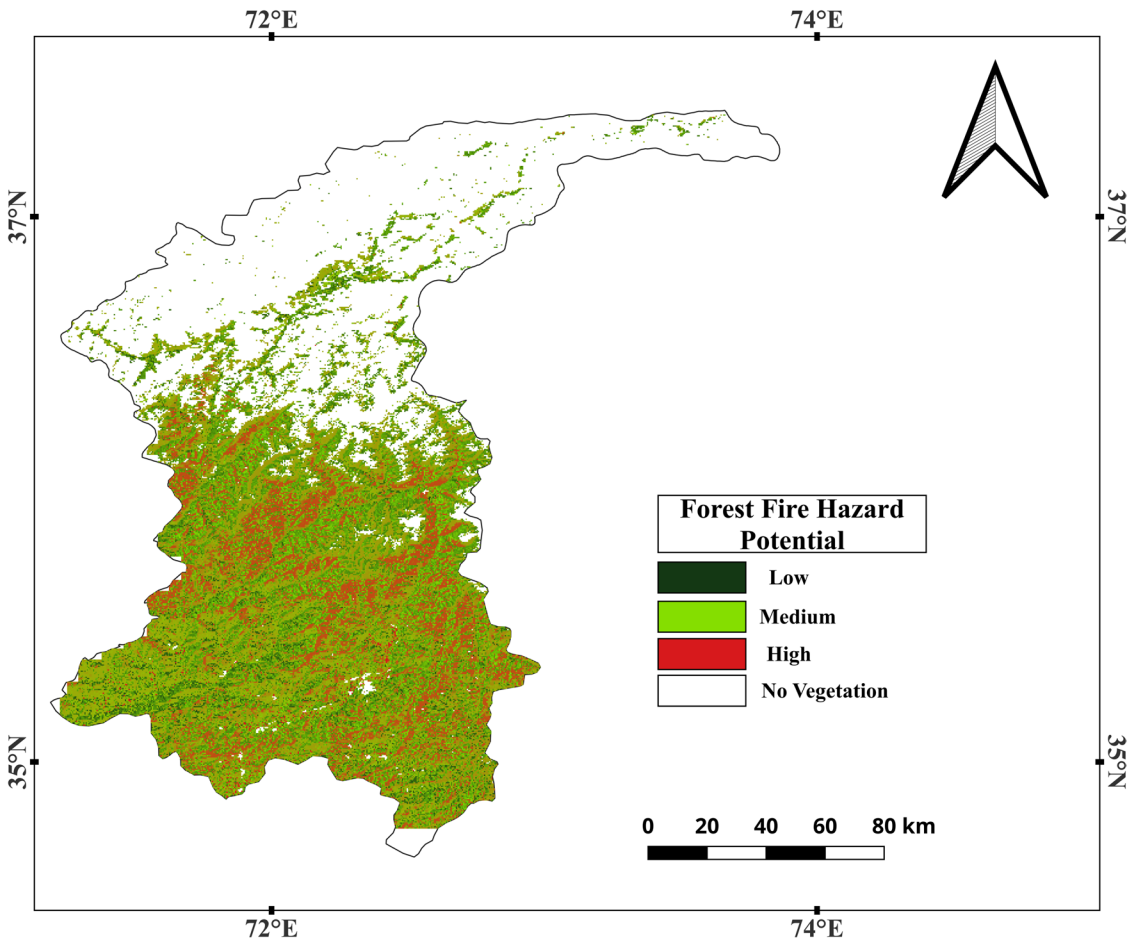

Fig. 7 Predicted forest fire probability map for the Malakand Division. The map categorizes fire risk into three classes: low probability (dark green), medium probability (light green), and high probability (red), with areas of no vegetation shown in white

management efforts across high-risk areas in the Malakand Division.

Discussion

The findings of this study reveal the dynamic nature of forest fires in the Malakand division over a 23-year period. Statistical analysis using the Mann-Kendall and Sen's slope tests identified an annual upward trend, with a Kendall's tau value of 0.323 and a p-value of 0.032, indicating statistical significance. Sen's slope estimation suggested an average annual increase of one fire event, with distinct periods of heightened activity in 2009, 2015, 2018, and 2022. However, no significant monthly trends were observed (p-value: 0.381). Sen's slope analysis estimated a potential average rise of approximately 2.29 monthly fires during peak seasons, particularly in the warmer and drier months of June and July. These seasonal patterns align with findings from other studies, where statistical techniques have revealed the cyclic nature of wildfire occurrences (Ahmed and Hassan 2023; Jain et al. 2017). Comparing these trends with European Mediterranean (EUMed) countries underscores broader climate influences on fire dynamics. While countries like Portugal and Spain have experienced increasing fire occurrences, others, such as Italy, have seen notable declines, reflecting regional variations in fire management practices and environmental conditions (Rodrigues et al. 2013). Spatial analysis using the Standard Deviation Ellipse (SDE) method demonstrated shifts in the central locations and dispersion of fire-prone zones over time. These changes suggest that land use and vegetation density are critical in influencing fire dynamics. The evolving patterns align with prior research identifying these factors as significant determinants in shaping wildfire behavior (Alexander 1985; Shao et al. 2023). The findings emphasize the complexity of wildfire prediction, particularly due to the spatial variability of factors contributing to fire ignition and spread.

Regarding multicollinearity, it reduces the precision of estimated quantities and the potential efficiency of the

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classification model (Hong et al. 2018). A reliable forest fire Probability map was developed by initially testing the likelihood of various links among the 34 fire-conditioning factors. Owing to strong linear associations among the conditioning factors, VIF, and tolerance values revealed that only 16 factors were preferred in the model construction phase. Before discussing the essential outcomes within each study, we first explore the factors influencing fire ignition occurrence. Several studies reviewed in this paper, including those referenced as (Chang et al. 2013; Vasilakos et al. 2009), assess the factors influencing fire occurrence and risk. According to (Oliveira et al. 2012), the factors and conditions influencing fire occurrences vary significantly across regions, mainly due to differing environmental and social conditions inherent to each country's unique ecosystem and studied area. In the Malakand division, the influence of human activity, topography, vegetation, and climate on the ignition of forest fires was assessed using an ML approach with a RF model. The RF model identified vegetation factors, particularly tree cover and Leaf Area Index (LAI), as major contributors to fire ignition. The critical role of vegetation management in fire prevention strategies is underscored by numerous studies (Ganteaume et al. 2013), suggesting that reducing fuel loads in high-risk areas could effectively diminish fire occurrence probabilities. Climatic factors also play an important role in stating the condition of fire fuel, as highlighted by (Atchley et al. 2021; Benali et al. 2017), and are of great importance for fire ignition (Varner et al. 2015). This study confirms the significant impact of climatic factors such as vapor pressure, soil temperature, and precipitation on fire ignition. Weather indices also correspond well with forest fire frequency locally and nationally (Jurečka et al. 2019). Among anthropogenic features, population density appeared as a critical variable in our study, while topographic factors showed a much lower ignition probability (Zhao et al. 2021).

The predictive modeling of forest fire incidents demonstrates that integrating various models through a multi-layer stacking ensemble approach significantly enhances prediction accuracy and reliability. Traditional single-model approaches, such as RF, SVM, or LR, often struggle with specific limitations, such as overfitting, lack of robustness to diverse data structures, or lower generalization capability. For instance, RF is highly effective but can be sensitive to noisy data, while SVM may face challenges in handling large datasets or complex, nonlinear relationships (Jain et al. 2020). The stacking ensemble approach addresses these shortcomings by combining the strengths of multiple algorithms in a structured multi-layer framework. This method reduces bias and variance while enhancing the model's ability to generalize across diverse data. The ensemble model in our study achieved an impressive accuracy of 96.24%, an F1 score of 96.17%, and a ROC AUC of 99.43%, showcasing its effectiveness in tackling the complexities of forest fire susceptibility prediction (Kim and Jeong 2022).

The superior performance of the ensemble model over individual models such as RF, XGBoost, SVM, LR, and KNN is consistent with recent findings in environmental modeling literature, where ensemble methods often provide more reliable predictions in ecological and meteorological contexts (Y.-S. Chang et al. 2020d Zhang et al. 2020). By combining multiple ML models to reduce variance or bias, the ensemble approach proved most effective in this study. It harnesses the strengths of various algorithms, aggregating their predictions to improve forecasting accuracy. Additionally, the Bayesian average can be employed to address regression problems or compute classification probabilities, further enhancing predictability and credibility (Derbentsev et al. 2021; Lieske et al. 2018). The robust precision and recall rates underscore the model's efficacy, particularly in real-world applications where prediction errors can have significant consequences (Sasikala and Theetchenya 2024; Sukmana et al. 2024). Given its demonstrated effectiveness, the ensemble model offers practical value beyond academic research, presenting a potent tool for fire management agencies tasked with risk assessment and disaster response planning. This aligns with the broader discourse on advancing predictive modeling techniques to enhance precision and responsiveness in environmental management (Jain et al. 2020; Li et al. 2023; Cao et al. 2024; Fan et al. 2024).

The methods employed in this study to develop the forest fire probability map and analyze past fire occurrences in the Malakand Division demonstrate promising potential for future applications. This research establishes a benchmark for accuracy and reliability in forest fire risk assessment by integrating single and ensemble ML models with advanced spatial analysis tools. However, the study is limited by the absence of real-time data integration, such as continuously updated satellite imagery and on-ground sensor networks, which could improve the precision and timeliness of predictions. Future research should prioritize the development of dynamic models that incorporate real-time data to enhance adaptability and responsiveness in forest fire management. Additionally, this study underscores the importance of interdisciplinary collaboration among data scientists, ecologists, and policymakers to establish effective forest management practices (Anees et al. 2022a, b; Akram et al. 2022; Andreevich et al. 2020; Aslam et al. 2022; Khan et al. 2024; Shobairi et al. 2022; Usoltsev et al. 2020, 2022). The findings have significant policy implications, encouraging the adoption of data-driven decision-making frameworks by governmental agencies for fire prevention and resource allocation. Future studies could explore the potential of deep learning models, which may provide even greater predictive power when combined with comprehensive environmental

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datasets. The methods developed here have global relevance and can be adapted to diverse ecological regions, inspiring researchers to replicate similar approaches in other parts of the world. Integrating satellite data with ground-based information presents an opportunity to transform current environmental risk management systems. This approach can play a pivotal role in tackling the escalating challenges of climate change and forest fire management by addressing limitations and leveraging future advancements.

Conclusion

This study developed a ML framework for forest fire prediction in the Malakand Division, achieving high accuracy and reliability using a multi-layer stacking ensemble model. The analysis identified tree cover, LAI, vapor pressure, and soil temperature as the most critical factors influencing fire ignition. A significant upward trend in annual fire occurrences was observed over 23 years, with high-risk periods concentrated in June, July, and November. The forest fire probability map highlights districts such as Shangla, Buner, and Dir as high-risk areas, providing essential insights for targeted fire management strategies. These findings not only enhance understanding of fire dynamics but also offer actionable tools for risk mitigation. Key recommendations include implementing targeted fire management strategies in identified high-risk zones, integrating the predictive framework with real-time monitoring systems for early warnings, and promoting sustainable land management practices to reduce fire-prone conditions. This study establishes a reliable and practical approach to forest fire prediction, bridging the gap between advanced ML techniques and real-world applications. The results contribute valuable tools for decision-making and highlight the potential for expanding this methodology to other fire-prone regions.

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Author contributions Fahad Shahzad: conceptualization, methodology, software, formal analysis, visualization, data curation, writing—original draft, investigation, validation, writing—review and editing. Kaleem Mehmood: visualization, writing—review and editing, Zhongke Feng: writing—review and editing, Supervision. Khadim Hussain: writing—review and editing. Ijlal Haidar: writing—review and editing. Shoaib Ahmad Anees: formal analysis, software, investigation, writing—review and editing. Sultan Muhammad: writing—review and editing. Jamshid Ali: writing—review and editing. Muhammad Adnan: writing—review and editing. Waseem Razzaq Khan: formal analysis, investigation, writing—review and editing.

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Data availability The authors confirm that the data links supporting the findings of this study are available within the article.

Declarations

Competing interests The authors declare no competing interests.

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